

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Sefaxersen, an Antisense Inhibitor of Complement Factor B, in Patients With Primary IgA Nephropathy at High Risk of Progression **Short Title/Acronym:** IMAGINATION **Protocol Number:** WA43966 / 2022-502102-32-00 **NCT Number:** NCT05797610 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Sefaxersen (R07434656) **Phase of Development:** Phase III **Medical Monitor:** Head, EU / International Pharmaceuticals Division, Hoffmann-La Roche

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and pharmacokinetics of sefaxersen (R07434656) compared to placebo in participants with primary IgA nephropathy (IgAN) who are at high risk of progressive kidney disease despite optimized supportive care. **Secondary Objectives:** To evaluate the long-term effect of sefaxersen on kidney function preservation, including estimated Glomerular Filtration Rate (eGFR) stabilization and sustained reduction in proteinuria.

2.2 Background & Rationale IgA nephropathy (IgAN) is the most widespread primary glomerular disease and a leading cause of chronic kidney disease and kidney failure. The complement alternative pathway plays a paramount role in the pathogenesis of IgAN, triggering downstream tissue injury and damage. To address the unmet need for targeted treatments that can halt progression in high-risk patients, this study investigates sefaxersen, a novel Antisense Oligonucleotide (ASO) designed to inhibit complement factor B. By suppressing this pathway, the study aims to reduce proteinuria and prevent long-term kidney function decline.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 (Sefaxersen vs. Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 428 participants **Number of Sites:** Approximately 201 locations worldwide **Estimated Study Start:** 08-Aug-2023 **Estimated Primary Completion:** 31-Aug-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men or women ≥ 18 years of age. 2. Confirmed diagnosis of primary IgA Nephropathy (IgAN) within the prior 10 years. 3. Urine Protein-to-Creatinine Ratio (UPCR) ≥ 1 gram per gram (g/g) or urine protein excretion ≥ 1 gram per day (with UPCR ≥ 0.8 g/g). 4. eGFR ≥ 20 mL/min/1.73 m². 5. Vaccination against *Neisseria meningitidis*, *Streptococcus pneumoniae*, and *Haemophilus influenzae* according to national recommendations.

4.2 Exclusion Criteria 1. Pregnancy or breastfeeding, or intention of becoming pregnant during the study or within 12 weeks after the final dose. 2. Histopathologic or other evidence of another autoimmune glomerular disease. 3. Presence of ≥ 50% crescents on kidney biopsy, sustained doubling of serum creatinine within 3 months prior to screening, or rapidly progressive glomerulonephritis. 4. History of kidney transplantation. 5. Glycated Hemoglobin (HbA1c) ≥ 6.5% or a clinical diagnosis of diabetes mellitus of any type. 6. Usage of Glucagon-like Peptide-1 (GLP-1)-based therapy within 90 days prior to screening.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Subcutaneous injection every 2 weeks for the first 3 doses, and then every 4 weeks. Route of Administration: Subcutaneous (SC) Duration of Treatment: Approximately 2 years (104/105 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Optimized supportive care (e.g., standard of care RAAS blockade). Prohibited: GLP-1 receptor agonists (within 90 days prior to screening or initiated during the study period), and other immunosuppressive therapies not permitted by the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, eGFR, 24-hr urine UPCR, Vaccinations
Baseline	Day 0	Randomization, First SC Dose, Baseline Labs
Interim	Every 2 to 12 weeks	Study drug administration, Safety Labs, UPCR/eGFR Assessments
End of Study	Week 105 (+ Follow-up)	Final Efficacy Assessment, Exit Interview, Safety follow-up 3 months (12 weeks) after final dose

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in Urine Protein-to-Creatinine Ratio (UPCR) at Week 37. **Measurement Method:** UPCR assessed via 24-hour urine collection and central laboratory analysis.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in estimated Glomerular Filtration Rate (eGFR) at Week 105. 2. Safety and tolerability profile (incidence and severity of AEs/SAEs). 3. Pharmacokinetics of sefaxersen (R07434656).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via clinical evaluation and laboratory testing during routine site visits scheduled every 2 to 12 weeks. **Serious Adverse Events (SAEs):** Standard reporting to the sponsor/regulatory agencies within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent DMC will review unblinded safety and efficacy data continuously to ensure participant safety and trial integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy evaluations; Safety Set for all participants who received at least one dose of the investigational product. **Sample Size Justification:** $N=428$, adequately powered to detect a statistically significant reduction in UPCR at Week 37 and a favorable stabilization in the eGFR slope at Week 105 compared to placebo. **Handling of Missing Data:** Repeated measures mixed-effect models (MMRM) and multiple imputation techniques for continuous laboratory variables like eGFR and UPCR.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to evaluate protocol adherence, source data verification, and drug accountability. **Audit Trail:** Utilizing a validated Electronic Data Capture (EDC) system with secure eCRF locking, automated data clarification forms, and comprehensive audit trails.

11. Study Identifiers & Governance

Primary ID: WA43966 **Registry Number:** NCT05797610 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** IMAGINATION **Official**

Regulatory Title: A Phase III, Multicenter, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Sefaxersen, an Antisense Inhibitor of Complement Factor B, in Patients With Primary IgA Nephropathy at High Risk of Progression

12. Clinical Framework (IgA Nephropathy Specific)

Primary Condition: Primary Immunoglobulin A (IgA) Nephropathy (IgAN) **Diagnosis Threshold:** UPCR ≥ 1 g/g and eGFR ≥ 20 mL/min/1.73 m² **Medical Methodology:** Optimized supportive care in combination with targeted Antisense Oligonucleotide (ASO) therapy **Optimization Target:** Proteinuria reduction and eGFR stabilization

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for IgAN augmented with investigational complement factor B inhibition. **Education Protocol (HCP):** Comprehensive protocol, safety reporting, and subcutaneous injection training during Site Initiation Visits. **Education Protocol (Patient):** Informed consent onboarding and disease education materials surrounding progressive kidney disease. **Quality Audit Mechanism:** Centralized laboratory sample tracking, and routine clinical monitoring of HCP adherence to the 1:1 blinded randomization and dosing schedule.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 105 weeks of treatment + 12 weeks of safety follow-up **Onsite Visit Cadence:** Every 2 to 12 weeks during the treatment phase **Remote Monitoring Frequency:** Intermittent remote assessments managed by the study coordinator/investigator. **Checklist Review Interval:** AEs, SAEs, and concomitant medication reviews conducted at every scheduled visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				